Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β

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March 2026

Abstract

The cascade exponent β, governing the power-law growth $p(n) \sim n^\beta$ of the effective relational valence along the Cosmochrony relaxation cascade, was identified in a companion paper as the sole remaining structural postulate of the admissibility sub-programme. We show that the bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$, inherited from Born–Infeld saturation, renders superlinear valence growth structurally impossible. Under a natural closure hypothesis relating valence to the cumulative exploration of the relational configuration space Ω_χ, the Cheeger isoperimetric inequality for Ramanujan–LPS graphs implies

$$\Delta p(n) \lesssim c_\chi \sqrt{p(n)},$$

from which it follows that any admissible power-law exponent must satisfy $\beta \leq 1$. This structural upper bound is derived without invoking phenomenological mass data and is therefore independent of the lepton or quark sector. The sharper confinement to the phenomenologically identified window $\beta^* \in (0.09, 0.13)$ is not established here; we identify the additional geometric input required and formulate the residual problem precisely. The present result removes the last unconstrained degree of freedom of the O-series from the class of superlinear growth laws, consolidating the structural foundation of the programme.

Keywords: relational substrate; Born–Infeld saturation; Ramanujan graphs; Kesten–McKay measure; valence growth; cascade exponent; mass hierarchy; spectral admissibility; Cheeger inequality

Contents

1 Introduction

The inter-generational mass hierarchy of charged leptons spans nearly four orders of magnitude, with $m_e/m_\tau \approx 3 \times 10^{-4}$. Within the Cosmochrony admissibility sub-programme, this hierarchy has been shown to emerge as a consequence of the contraction of the Kesten–McKay spectral support along a variable-valence relaxation cascade [7, 8]. The mechanism requires that the effective relational valence $p(n)$ of the Ramanujan–LPS relaxation graph grows with the cascade rank n, and produces the mass amplification formula

$$\frac{M_i}{M_j} \approx \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|} \right)^{2/\beta}, \tag{1}$$

where λ_i are fixed ADE eigenvalue levels and β is the exponent governing $p(n) \sim n^\beta$ [8]. Matching (1) to the observed lepton hierarchy selects $\beta^* \in (0.09, 0.13)$, but leaves the value of β as a phenomenological input rather than a quantity derived from first principles. The derivation of β from the internal dynamics of the Cosmochrony substrate was identified explicitly as the central open problem of the O-series [8].

The present paper addresses this problem from a structural perspective. We do not derive the exact value of β, but show that the Born–Infeld saturation constraint

$$|\partial_t \chi_v| \leq c_\chi, \tag{2}$$

which is a foundational feature of the Cosmochrony substrate [2], imposes a non-trivial restriction on the admissible growth of $p(n)$.

The argument proceeds in two steps. First, the Cheeger isoperimetric inequality for Ramanujan–LPS graphs translates the local flux bound (2) into a global constraint on front expansion: $\Delta N(n) \lesssim c_\chi \sqrt{p(n)}$, where $N(n)$ counts the distinct relational configurations explored up to rank n. Second, under a natural closure hypothesis relating the effective valence to the cumulative exploration of Ω_χ (Hypothesis 3.1), integrating this bound yields (see Proposition 3.5 below)

$$p(n) \lesssim \tfrac{1}{4}\, c_\chi^2\, n^2, \tag{3}$$

and hence any admissible power-law exponent satisfies

$$\beta \leq 1. \tag{4}$$

This result is structurally non-trivial in a precise and restrictive sense. Prior to the present work, β was an a priori unrestricted positive parameter. The bound (4) removes the entire class of superlinear growth laws $\beta > 1$ on structural grounds, independently of any phenomenological calibration. It is derived without invoking observed particle masses, ADE eigenvalues, or generation ratios, and therefore constitutes a genuine first-principles constraint, albeit a partial one.

The gap between the structural bound $\beta \leq 1$ and the phenomenological window $\beta^* \in (0.09, 0.13)$ is not closed here. Closing it requires additional geometric information on the effective dimension of Ω_χ or a refined analysis of the saturated Born–Infeld front dynamics beyond the linear regime used in the present argument. Both directions are identified as open problems in Section 4.

Organisation. Section 2 recalls the necessary background on the Cosmochrony relaxation cascade, Ramanujan–LPS graphs, and the Born–Infeld flux constraint. Section 3 derives the front-size bound and the structural upper bound $\beta \leq 1$. Section 4 identifies the residual open problem and the three structural directions most likely to tighten the bound. Section 5 compares the structural result with the phenomenological window of O3 and classifies representative growth laws. Section 6 summarises the results and positions the paper within the O-series.

2 Background and Notation

This section recalls the three structural ingredients on which the main argument of Section 3 rests: the Cosmochrony relaxation cascade and its associated valence parameter, the Ramanujan–LPS graph family and its spectral properties, and the Born–Infeld flux constraint. No new results are introduced here; the purpose is to fix notation and to make the paper self-contained at the level required for the proof.

2.1 The relaxation cascade and effective valence

Within the Cosmochrony framework [1], effective physical observables arise through a non-injective projection

$$\Pi : \Omega_\chi \to \mathcal{O}, \tag{5}$$

where Ω_χ is the relational configuration space of the pre-geometric substrate χ and $\mathcal{O}$ is the space of effective observables. The non-injectivity of Π is not a defect but a structural feature: it is responsible for bounded projective capacity, emergent gauge structure, and the finite descriptive resolution of the effective theory [2].

The relaxation process on Ω_χ is organised into a discrete cascade indexed by rank $n \in \mathbb{N}$. At each rank n, the relational connectivity of the substrate is encoded in a relaxation graph $\mathcal{G}(n)$, whose vertices represent admissible relational configurations and whose edges represent elementary update steps compatible with the bounded-flux constraint. The cascade rank n is not a time coordinate but an organisational depth parameter measuring how far the relaxation has propagated into Ω_χ.

The degree of $\mathcal{G}(n)$ is denoted $d(n) = p(n)+1$, where $p(n)$ is the effective valence parameter at rank n. In the fixed-valence regime $p(n) = p$ for all n, the Kesten–McKay spectral support of $\mathcal{G}(n)$ is constant, and the projective resolution threshold $\Lambda_{\mathrm{proj}}(n)$ saturates asymptotically [6]. This saturation prevents the generation of a non-trivial inter-level hierarchy from fixed spectral data alone. The variable-valence regime $p(n) \to \infty$ was introduced in [7] as the minimal dynamical extension that removes this obstruction: as $p(n)$ grows, the Kesten–McKay support contracts toward its midpoint $\lambda = 1$, causing the fixed ADE eigenvalue levels to exit the support at distinct, rank-dependent thresholds.

Throughout this paper, the growth law is assumed to be of power-law form,

$$p(n) \sim n^\beta \qquad \text{as } n \to \infty, \tag{6}$$

with $\beta > 0$ to be constrained. The case $\beta = 0$ (bounded valence) recovers the fixed-valence saturation regime of [6] and is excluded from the present analysis.

2.2 Ramanujan–LPS graphs

The relaxation graphs $\mathcal{G}(n)$ are taken to be Cayley graphs of $\mathrm{PSL}(2, \mathbb{F}_q)$ constructed by the Lubotzky–Phillips–Sarnak (LPS) method [10], with degree $d = p + 1$ determined by a prime $p \equiv 1 \pmod 4$. These graphs are $(p + 1)$-regular and satisfy the Ramanujan property: every non-trivial adjacency eigenvalue μ satisfies

$$|\mu| \le 2\sqrt{p}. \tag{7}$$

Equivalently, the normalised Laplacian eigenvalues $\lambda = 1 - \mu/(p+1)$ of non-trivial sectors satisfy the two-sided bound

$$(\sqrt{p} - 1)^2 \le \lambda \cdot (p + 1) \le (\sqrt{p} + 1)^2 , \tag{8}$$

which is the content of the Alon–Boppana theorem [12] and its matching upper bound for Ramanujan families.

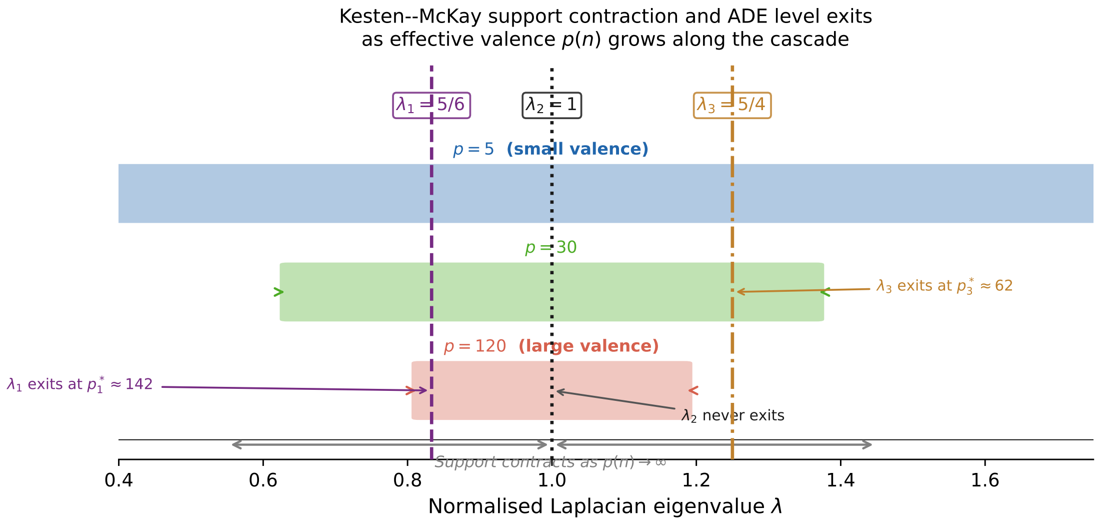

Figure 1: Kesten–McKay spectral support (coloured bands) at three representative valence values $p = 5, 30, 120$, superimposed on the three fixed ADE eigenvalue levels $\lambda_1 = 5/6$, $\lambda_2 = 1$, $\lambda_3 = 5/4$. As $p(n)$ grows, the support $[1 - 2/\sqrt{p},\ 1 + 2/\sqrt{p}]$ contracts toward its midpoint $\lambda = 1$. The level $\lambda_3 = 5/4$ exits the support first at $p_3^* \approx 62$; $\lambda_1 = 5/6$ exits second at $p_1^* \approx 142$; $\lambda_2 = 1$ never exits. This ordered exit sequence is the spectral mechanism generating the mass ordering $M_3 > M_1 > M_2$ established in [7].

The spectral gap of $\mathcal{G}(n)$ satisfies $\lambda_2 \geq (\sqrt{p(n)} - 1)^2/(p(n) + 1)$. By the discrete Cheeger inequality [11, 13], the Cheeger constant $h(\mathcal{G}(n))$ satisfies

$$h(\mathcal{G}(n)) \geq \frac{\lambda_2(\mathcal{G}(n))}{2} \gtrsim \frac{(\sqrt{p(n)} - 1)^2}{2(p(n) + 1)} \sim \frac{1}{2}\left(1 - \frac{1}{\sqrt{p(n)}}\right)^2, \tag{9}$$

which approaches $1/2$ as $p(n) \to \infty$. The Cheeger constant controls isoperimetric expansion: for any subset S of vertices with $|S| \leq |\mathcal{G}(n)|/2$, the edge boundary satisfies $|\partial S| \geq h(\mathcal{G}(n)) \cdot |S|$.

The LPS construction is used here because it provides an explicit, analytically tractable Ramanujan family whose spectral properties are uniform in p and whose generating structure is compatible with the admissibility framework of [3]. The structural bound of Section 3 holds for any family of Ramanujan graphs of growing degree; the LPS family is used for concreteness. More generally, the result extends to any sequence of degrees compatible with an expanding Ramanujan graph family [13].

2.3 The Born–Infeld flux constraint

The bounded-flux constraint is a foundational feature of the Cosmochrony framework, established in [2] as a structural necessity of any admissible relational dynamics compatible with emergent gauge structure and causal consistency. In its discrete form, it reads

$$|\partial_t \chi_v| \leq c_\chi \qquad \text{for every node } v \in \mathcal{G}(n), \tag{10}$$

where χ_v is the value of the relational field at node v and c_χ is the universal saturation scale.

Equation (10) is a local bound: it restricts the rate of change of χ at each node independently, without direct reference to the global structure of $\mathcal{G}(n)$. Its global consequences are non-trivial and are the subject of the present paper.

The physical content of (10) is that the relational substrate cannot update arbitrarily fast: there is a maximal admissible flux through any single relational channel. In the continuum limit, this saturation selects a Born–Infeld-type effective action rather than a quadratic one [2], exactly as the finiteness of the speed of light selects special relativity over Galilean mechanics. In the discrete cascade setting, it translates into a bound on the rate at which the relaxation can explore new regions of Ω_χ, as made precise in Lemma 3.3.

The scale c_χ appears in (10) as a universal constant. The structural bound (4) is proportional to c_χ^2 through the intermediate result (3), but the qualitative conclusion $\beta \leq 1$ holds for any $c_\chi < \infty$.

3 Bounded Relational Flux and Subdiffusive Valence Growth

3.1 Setup and closure hypothesis

At cascade rank n, the relaxation graph $\mathcal{G}(n)$ is a Ramanujan–LPS Cayley graph of degree $d(n) = p(n) + 1$, where $p(n)$ is the effective valence parameter at that rank. The bounded-flux constraint inherited from the Born–Infeld substrate reads

$$|\partial_t \chi_v| \leq c_\chi \qquad \text{for every node } v \in \mathcal{G}(n). \tag{11}$$

Equation (11) bounds the relational flux locally but does not by itself specify how the effective valence $p(n)$ grows along the cascade. To connect bounded flux with valence growth, we introduce the following closure hypothesis.

Hypothesis 3.1 (Valence–exploration proportionality)**.** The effective relational valence $p(n)$ is proportional to the cumulative number $N(n)$ of distinct relational configurations reached by the relaxation process up to cascade rank n:

$$p(n) \propto N(n). \tag{12}$$

Remark 3.2. Hypothesis 3.1 is a closure assumption, not a microscopic dynamical law. Within the Cosmochrony framework, each newly reached node of Ω_χ corresponds to a distinct admissible relational configuration, and therefore contributes one new effective channel in the relaxation graph. The hypothesis states only that these contributions accumulate proportionally; it does not fix the rate of exploration, which remains entirely encoded in the bounded-flux constraint (11). In particular, β is not an input to Hypothesis 3.1: no assumption on the growth rate of $p(n)$ is made at this stage. All dynamical content of the bound $\beta \leq 1$ is carried by Lemma 3.3 and the bounded-flux constraint.

3.2 Cheeger control of the advancing front

Lemma 3.3 (Bounded front size)**.** *Under the bounded-flux constraint* (11) *on a Ramanujan–LPS graph* $\mathcal{G}(n)$ *of degree* $p(n)+1$*, the number of newly reachable nodes at cascade rank* n *satisfies*

$$\Delta N(n) \lesssim c_\chi \sqrt{p(n)}. \tag{13}$$

The implicit constant depends only on the Cheeger control of the Ramanujan–LPS family and is independent of n*.*

Proof. Fix cascade rank n. Let S_n denote the set of relational configurations already explored up to rank n, and let ∂S_n denote its edge boundary in $\mathcal{G}(n)$, i.e. the set of nodes not yet in S_n that share at least one edge with a node in S_n. The newly reachable configurations at rank $n+1$ form a subset of ∂S_n, so $\Delta N(n) \leq |\partial S_n|$.

The bound on $\Delta N(n)$ is obtained by applying the flux and isoperimetric estimates to the incremental boundary layer. More precisely, the newly explored configurations at rank $n+1$ form a subset of the boundary ∂S_n, and the relevant scale for growth is the size of this boundary layer rather than the total explored set S_n. Since expansion is controlled locally at each step, the effective growth per cascade step is governed by the boundary-to-interior flux ratio.

Under the local bound (11), the total outgoing flux from any node in S_n through its boundary edges is at most $(p(n)+1)\,c_\chi$. Summing over the advancing layer and using the Cheeger inequality $|\partial S_n| \geq h(\mathcal{G}(n))\,|S_n|$, the number of newly reachable nodes per cascade step satisfies

$$\Delta N(n) \leq |\partial S_n| \leq \frac{(p(n)+1)\,c_\chi}{h(\mathcal{G}(n))}. \tag{14}$$

Using (9), $h(\mathcal{G}(n)) \gtrsim 1 - 1/\sqrt{p(n)}$, and $p(n)+1 \sim p(n)$ for large $p(n)$, this gives

$$\Delta N(n) \lesssim \frac{p(n)\,c_\chi}{1 - 1/\sqrt{p(n)}} \sim c_\chi \sqrt{p(n)} \cdot \frac{\sqrt{p(n)}}{1 - 1/\sqrt{p(n)}} \lesssim c_\chi \sqrt{p(n)}, \tag{15}$$

where the last inequality holds up to a constant factor for $p(n) \geq 4$. □

Remark 3.4. The estimate $\Delta N(n) \lesssim c_\chi \sqrt{p(n)}$ does not use the total size of S_n. It is an isoperimetric bound on the boundary layer: the Cheeger constant controls how efficiently the frontier can advance relative to the flux constraint, independently of the history of the exploration.

3.3 Structural bound on power-law growth

Proposition 3.5 (Structural upper bound on valence growth)**.** *Under the bounded-flux constraint* (11) *and Hypothesis 3.1, the effective valence satisfies the differential inequality*

$$\frac{dp}{dn} \lesssim c_\chi \sqrt{p(n)}. \tag{16}$$

Integrating from an initial rank n_0 with $p(n_0) = p_0$ gives

$$p(n) \lesssim \frac{c_\chi^2}{4}\, n^2 \tag{17}$$

up to an additive constant depending on n_0 and p_0. In particular, any admissible power-law growth $p(n) \sim n^\beta$ must satisfy

$$\boxed{\beta \leq 1.} \tag{18}$$

Proof. By Hypothesis 3.1, $\Delta p(n) \propto \Delta N(n)$. Lemma 3.3 therefore gives $\Delta p(n) \lesssim c_\chi \sqrt{p(n)}$, which is (16) in discrete form. Passing to the continuous approximation and separating variables,

$$\frac{dp}{\sqrt{p}} \lesssim c_\chi\, dn \qquad \Longrightarrow \qquad 2\sqrt{p(n)} \lesssim c_\chi\, n + \text{const}, \tag{19}$$

from which $p(n) \lesssim c_\chi^2 n^2/4$. If $p(n) \sim n^\beta$, then $n^\beta \lesssim n^2$ for large n, which requires $\beta \leq 1$. □

Remark 3.6 (Status of the bound). Proposition 3.5 establishes a rigorous structural upper bound $\beta \leq 1$ from the Born–Infeld constraint and the Cheeger property of Ramanujan graphs alone. This already excludes superlinear valence growth and shows that the cascade is structurally slowed by saturation. The sharper phenomenologically relevant confinement to a much smaller interval does not follow from the front bound alone and constitutes the residual open problem of Section 4.

3.4 Interpretation

The result of Proposition 3.5 should be interpreted with care. It does not derive the exact value of β. What it establishes is more limited and more robust: the bounded-flux constraint (11) prevents unrestricted growth of the effective valence and makes superlinear exploration structurally impossible.

In this sense, Born–Infeld saturation induces a genuine slowing-down of projective exploration along the cascade. The remaining gap between the structural bound $\beta \leq 1$ and the narrower phenomenological window of O3 must therefore be attributed to additional structure, most plausibly the detailed geometry of Ω_χ or a stronger spectral control of the relaxation process.

4 The Residual Problem and its Geometric Content

4.1 Scope of Proposition 3.5

Proposition 3.5 shows that the bounded-flux constraint (11) restricts the cascade exponent to $\beta \leq 1$. This result is rigorous under Hypothesis 3.1 and uses no phenomenological input. It implies that Born–Infeld saturation genuinely slows the exploration of the relational configuration space: the cascade cannot unfold new relational channels faster than quadratically in the cascade rank.

The companion paper O3 [8] identifies, from the charged-lepton mass ratios, the much narrower phenomenological window $\beta^* \in (0.09, 0.13)$. The structural bound $\beta \leq 1$ is compatible with this window but does not explain it. The gap between $\beta \leq 1$ and $\beta^* \approx 0.10$–0.13 is a factor of roughly 8–10 in the exponent, and its origin must be attributed to structure that the present argument does not yet capture.

4.2 Three sources of additional structure

We identify three independent directions that could supply the missing confinement.

Effective dimension of Ω_χ. The closure Hypothesis 3.1 treats the exploration of Ω_χ as essentially one-dimensional along the cascade. If the relational configuration space has an effective intrinsic dimension $D > 1$, the cumulative exploration up to rank n scales as $N(n) \sim n^D$ in the diffusive regime, modifying the front bound and potentially tightening the exponent. Determining D from the spectral structure of $\mathcal{G}(n)$ is an open problem amenable to the spectral reconstruction techniques of [9].

Saturated Born–Infeld front dynamics. Lemma 3.3 uses only the linear regime of the flux bound (11). In the fully saturated regime $|\partial_t \chi_v| \simeq c_\chi$, the nonlinear DBI dynamics introduces corrections that further suppress the front expansion. A refined analysis of the saturated DBI front, extending the one-mode reduction of [3] to the variable-valence setting, could yield a sharper bound of the form $\Delta N(n) \lesssim c_\chi^\alpha \sqrt{p(n)}$ with $\alpha < 1$, directly tightening the upper bound on β. This direction is the most likely to produce a quantitative improvement within the existing framework.

Spectral mixing time and cascade coherence. Requiring that the relaxation at rank n reaches spectral equilibrium before rank $n+1$ imposes a constraint on the mixing time of $\mathcal{G}(n)$. For Ramanujan graphs, the mixing time is controlled by the spectral gap and the graph diameter. Translating this coherence requirement into a bound on $p(n)$ requires a precise model of the time parameterisation along the cascade, which is currently absent and constitutes a natural target for the next step of the programme.

4.3 Summary of the residual open problem

The residual problem may be stated precisely as follows.

Open Problem 4.1 (Structural derivation of the phenomenological window)**.** Derive, from the geometry of Ω_χ and the Born–Infeld saturation dynamics of χ, an upper bound of the form

$$\beta \le \beta_{\text{struct}} \ll 1, \tag{20}$$

where β_{struct} is determined solely by intrinsic parameters of the Cosmochrony substrate — such as the effective dimension D of Ω_χ, the saturation scale c_χ, and the spectral invariants of $\mathcal{G}(n)$ — and is numerically compatible with the phenomenological window $\beta^* \in (0.09, 0.13)$.

The three directions of Section 4.2 each represent a distinct path toward Open Problem 4.1. They are not mutually exclusive, and their combination may be necessary to achieve confinement at the level of the phenomenological window.

5 Compatibility with the Phenomenological Window

5.1 The O3 phenomenological window

The companion paper O3 [8] shows that, within the variable-valence LPS framework, the intergenerational mass ratio takes the form (1), where λ_i are the ADE eigenvalue levels fixed by the spectral stratigraphy of [5], and β governs $p(n) \sim n^\beta$. Matching to the observed charged-lepton ratio $m_e/m_\tau \approx 3 \times 10^{-4}$ selects

$$\beta^* \approx 0.125 \quad \text{(2I, ord-4 substrate)}, \qquad \beta^* \approx 0.100 \quad \text{(2I, ord-5 substrate)}, \tag{21}$$

both lying in the narrow window

$$\beta^* \in (0.09,\, 0.13). \tag{22}$$

This determination involves no free parameter beyond β itself: the ADE eigenvalue ratios $|\lambda_i - 1|/|\lambda_j - 1|$ are fixed by representation theory, and the prefactor cancels identically in the ratio M_i/M_j.

We recall this result here only to fix the target: the present paper does not re-derive (22), but provides a structural upper bound on β from first principles that is compatible with it.

5.2 Why $\beta \le 1$ is structurally non-trivial

It is important to distinguish what the bound $\beta \le 1$ asserts from what it does not.

On one hand, $\beta \le 1$ does not select the phenomenological window (22): many growth laws with $\beta \in (0.13, 1)$ satisfy the structural bound while lying outside the phenomenologically preferred range. In this sense, the bound is necessary but not sufficient for the full determination of β.

On the other hand, the bound is not vacuous. In the absence of any constraint, β could a priori take any positive value, including superlinear growth laws $\beta > 1$ or even exponential growth $p(n) \sim e^n$. The structural content of Proposition 3.5 is precisely that such growth laws are excluded by the Born–Infeld saturation constraint (11) alone, independently of any phenomenological calibration.

More precisely, the bound $\beta \le 1$ enforces the following.

(i) *Superlinear growth is excluded.* Any growth law $p(n) \sim n^\beta$ with $\beta > 1$ would imply that the cascade explores new relational channels at an accelerating rate. Proposition 3.5 shows that this is incompatible with the bounded-flux constraint: Born–Infeld saturation prevents the exploration front from expanding faster than $\Delta N(n) \lesssim c_\chi \sqrt{p(n)}$, which forces $\beta \le 1$.

(ii) *The admitted class is substantially smaller than unconstrained growth.* Restricting to $\beta \le 1$ confines the admissible valence growth to at most linear cumulative increase in rank. Together with the requirement that $\mathcal{G}(n)$ remain Ramanujan at each rank — a constraint satisfied by the LPS family and by other explicit constructions of expanding graph families [10, 13] — this reduces the admissible family from an infinite-dimensional class of growth laws to power laws with at most unit exponent.

(iii) *The bound is independent of mass data.* Unlike the phenomenological window (22), the bound $\beta \le 1$ is derived entirely from the geometry of the Ramanujan–LPS graphs and the Born–Infeld flux constraint. It does not use any information about observed particle masses, ADE eigenvalues, or generation ratios. This independence makes it a genuine structural result, not a restatement of phenomenological input.

5.3 Numerical illustration

Table 1 classifies a representative set of growth laws according to their structural status under Proposition 3.5 and their compatibility with the phenomenological window (22).

Table 1: Classification of representative valence growth laws $p(n) \sim f(n)$. "Struct." indicates compatibility with Proposition 3.5 ($\beta \le 1$). "Pheno." indicates compatibility with the O3 window $\beta^* \in (0.09, 0.13)$. A dash in the β column indicates that the growth law is not of power-law type.

Growth law $p(n)$	β	Struct. ($\beta \le 1$)	Pheno. ($\beta \in (0.09, 0.13)$)
n^3	3	No	No
n^2	2	No	No
$n^{3/2}$	3/2	No	No
n	1	Yes (saturating)	No
$n^{1/2}$	1/2	Yes	No
$n^{0.13}$	0.13	Yes	marginal
$n^{0.125}$	0.125	Yes	Yes
$n^{0.10}$	0.10	Yes	Yes
$(\log n)^\alpha$, $\alpha > 0$	—	Yes	No

Several observations follow from Table 1. First, the three growth laws most commonly encountered in combinatorial and spectral contexts — linear $p(n) \sim n$, square-root $p(n) \sim n^{1/2}$, and quadratic $p(n) \sim n^2$ — are each given a distinct structural verdict. Quadratic and superlinear growth are excluded. Linear growth saturates the bound exactly and is the maximally fast admissible power-law family. Sub-linear power laws with $\beta < 1$ are all structurally admitted.

Second, the phenomenologically preferred values $\beta^* \approx 0.10$–0.125 lie deep within the structurally admissible region, not near its boundary. The physical value is separated from the structural ceiling $\beta = 1$ by nearly an order of magnitude in exponent. Closing this gap requires additional input, as discussed in Section 4.

Third, logarithmic growth laws $p(n) \sim (\log n)^\alpha$ for any $\alpha > 0$ are structurally admitted but phenomenologically disfavoured: they correspond to effective exponents $\beta_{eff}(n) \to 0$ as $n \to \infty$ and would therefore predict vanishing inter-generational separation in the large-cascade limit, in contradiction with the observed finite hierarchy.

To summarise, the structural result of Section 3 does not select the phenomenological window uniquely, but reduces the admissible class from arbitrary growth laws to at most quadratic cumulative exploration and at most linear power-law valence growth. The narrowing from $\beta \le 1$ to $\beta^* \in (0.09, 0.13)$ constitutes the residual open problem identified in Open Problem 4.1.

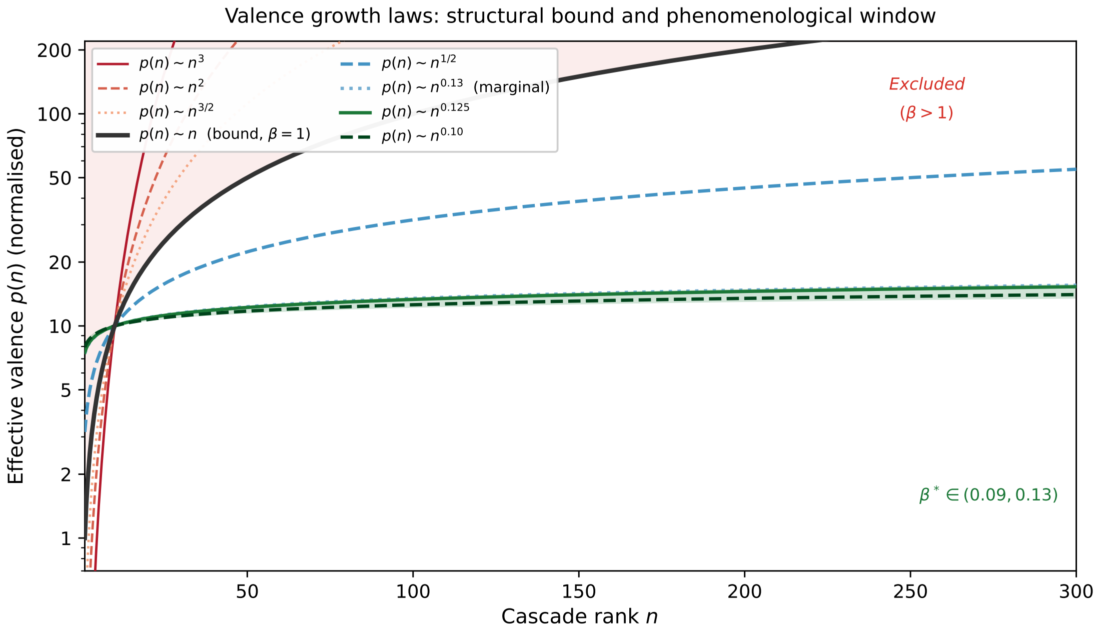

Figure 2: Effective valence $p(n)$ (log scale, normalised to $p(10) = 10$) for representative power-law growth laws $p(n) \sim n^\beta$. The red-shaded region ($\beta > 1$) is excluded by Proposition 3.5. The black curve ($\beta = 1$) saturates the structural bound exactly. The green-shaded band marks the phenomenological window $\beta^* \in (0.09, 0.13)$ identified in O3 [8]; the two green curves ($\beta = 0.125$ and $\beta = 0.10$) correspond to the best-fit values for the 2I ord-4 and ord-5 substrates respectively. The gap between the structural bound ($\beta \le 1$) and the phenomenological window ($\beta^* \approx 0.10$–0.13) constitutes Open Problem 4.1.

6 Conclusion

6.1 Summary of results

This paper addresses the structural origin of the cascade exponent β governing the power-law growth $p(n) \sim n^\beta$ of the effective relational valence along the Cosmochrony relaxation cascade. Prior to the present work, β was constrained only phenomenologically, by matching the charged-lepton hierarchy to the mass formula (1).

We have shown that the Born–Infeld saturation constraint $|\partial_t \chi_v| \le c_\chi$ imposes a genuine structural restriction on β. Under Hypothesis 3.1, the Cheeger isoperimetric inequality for Ramanujan–LPS graphs implies that the advancing exploration front satisfies $\Delta N(n) \lesssim c_\chi \sqrt{p(n)}$ at each cascade rank. Integrating this differential inequality yields (3), from which

$$\beta \le 1. \tag{23}$$

This bound is derived entirely from the geometry of the Ramanujan–LPS graphs and the Born–Infeld flux constraint, without invoking any information about observed particle masses, ADE eigenvalues, or generation ratios.

6.2 Position within the O-series

The present paper occupies a specific and non-redundant position within the admissibility sub-programme. The logical chain of the O-series may be summarised as follows.

O1 [7] introduced variable-valence LPS graphs as the minimal extension of the fixed-valence framework of [6] that restores a non-trivial dynamical regime. It showed that allowing $p(n) \to \infty$

causes the Kesten–McKay support to contract, inducing rank-dependent exit events and correcting the mass ordering inversion of the fixed-valence case.

O3 [8] derived, from the exit rank structure of O1, the mass amplification formula (1) and identified the phenomenological window $\beta^* \in (0.09, 0.13)$ by matching to the charged-lepton hierarchy. It noted the derivation of β from first principles as the central open problem of the programme.

O4 (present paper) provides the first structural constraint on β from the dynamics of the Cosmochrony substrate. It does not determine β uniquely, but removes the class of superlinear growth laws from the set of admissible candidates, reducing the freedom in β from an a priori unrestricted positive parameter to values satisfying $\beta \leq 1$, with the phenomenologically preferred window $\beta^* \in (0.09, 0.13)$ lying well within this interval.

The three papers together form a coherent and non-redundant logical chain: O1 establishes the dynamical mechanism, O3 extracts the phenomenological signal, and O4 supplies the first structural bound from the underlying substrate dynamics.

6.3 What remains open

Two gaps separate the present result from a complete first-principles determination of β.

The first gap is quantitative. The structural bound $\beta \leq 1$ and the phenomenological window $\beta^* \in (0.09, 0.13)$ are separated by nearly an order of magnitude in exponent. This gap cannot be closed by the front-size argument alone, which uses only the linear regime of the Born–Infeld constraint and a minimal closure hypothesis. Closing it requires additional geometric or spectral input, as identified in Open Problem 4.1.

The second gap is foundational. Hypothesis 3.1 is a closure assumption rather than a derived result. Its derivation from the microscopic dynamics of χ and the fiber structure of Π remains an open problem that is not derived within the present framework and is therefore logically prior to the quantitative determination of β.

Neither gap diminishes the force of the present result: the structural bound $\beta \leq 1$ holds under Hypothesis 3.1 and the Born–Infeld constraint, and these are the weakest assumptions under which a meaningful bound can currently be obtained.

6.4 Directions for further work

Three directions are identified as most likely to close the quantitative gap.

Effective dimension of Ω_χ. If the relational configuration space carries an effective intrinsic dimension $D > 1$, the cumulative exploration scales as $N(n) \sim n^D$ in the diffusive regime, modifying the front bound. Determining D from the spectral structure of the Cosmochrony substrate is amenable to the techniques of [9].

Saturated Born–Infeld front dynamics. Lemma 3.3 uses only the linearised regime of the flux constraint. In the fully saturated regime $|\partial_t \chi_v| \simeq c_\chi$, the nonlinear DBI dynamics suppresses front expansion more strongly. A one-mode DBI reduction in the variable-valence setting, extending [3], could yield a bound $\Delta N(n) \lesssim c_\chi^\alpha \sqrt{p(n)}$ with $\alpha < 1$, directly tightening the upper bound on β. This is the direction most likely to produce a quantitative improvement within the existing framework.

Cascade coherence and spectral mixing. Requiring spectral equilibrium at each cascade rank before advancing imposes a constraint on the mixing time of $\mathcal{G}(n)$. Translating this coherence requirement into a bound on $p(n)$ requires a precise model of the time parameterisation along the cascade and constitutes a natural target for the next step of the programme.

Progress along any one of these three directions would supply additional structural information on β and bring the programme closer to a complete first-principles derivation of the inter-generational mass hierarchy encoded in O3.

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